

Non-Complete Set Coverings for Higher Order Threshold Implementations

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Abstract

Side-channel attacks (SCAs) represent an important threat for the implementation of cryptographic algorithms. These attacks exploit the information leakage found in the physical magnitudes of hardware devices (e.g. current draw, electromagnetic emanation). Threshold Implementations (TIs) aim to mitigate SCAs by implementing a modified version of the algorithm that operates over randomized shares of its input and intermediate values. This strategy relies on the possibility of splitting the algorithm to be protected into sub-functions that satisfy certain properties about their dependence structure on the randomized shares. Non-complete set coverings (NCSCs) are combinatorial objects that can provide this dependence structure and guide the design of TIs. Given the desired order of protection d and the algebraic degree t of the functions to be implemented, for an NCSC to be useful, its cardinality r should be small and similar to the number of input shares s .

This work contributes to the study of NCSCs for efficient TIs by finding smaller coverings and proving novel theoretical bounds on their cardinality. We present a new NCSC for the case $t = 3, d = 2$ that is optimal and NCSCs for the cases $t = 3, d = 3$ and $t = 4, d = 2$ whose sizes are close to the lower bounds. We also present new combinatorial properties of these coverings and an algorithm for the search of small NCSCs.

Keywords: High-Order Masking, Non-Complete Set Coverings, Side channel attacks, Threshold Implementations

1 Introduction

Physical implementations of cryptographic algorithms must provide protection against Side Channel Attacks (SCAs). These threats exploit the information leakage found in

the physical magnitudes of a device, which are produced when it performs some sensitive tasks (see [1] for further details). The value of these magnitudes at each instant during the execution of the algorithm may be represented by a probability distribution. Kocher et al. [2] showed how the relation between the probability distribution of power consumption and the value of a specific intermediate bit can be leveraged to reveal the secret key of an encryption algorithm. Since then, successful attacks have been reported for a wide range of commonly used cryptographic algorithms, also relying on more sophisticated methods.

1.1 Masking techniques

Masking was one of the first techniques proposed to disrupt the relationship between leakage and sensitive information [3, 4]. It consists of splitting the algorithm into Boolean functions and for each one implementing an equivalent Boolean function that operates on a blinded representation of the inputs. For this, the input bit array (x) is randomly split into s bit arrays (x_i) of the same length, called *shares*. In Boolean masking, the bitwise sum of the shares retrieves the original input bit array as

$$x = \bigoplus_{i=1}^s x_i. \quad (1)$$

The sharing is done in such a way that the unmasked value is independent of the values of any $s - 1$ shares (*uniform sharing*). The function to be masked (f) is also split into r sub-functions ($\tilde{f}_i$) which operate over the shares instead of over the original input. This is done in a way that the outputs of the sub-functions are a sharing of the output of the original function when applied over the original input value. If $y = f(x)$, then $y_i = \tilde{f}_i(x_1, x_2, \dots, x_s)$ and

$$y = \bigoplus_{i=1}^r y_i. \quad (2)$$

As such, the desired output can be obtained by aggregating the outputs of the sub-functions. In the context of masking, we will refer to s as the number of input shares and to r as the number of sub-functions or output shares.

The aim of masking is that the relationship between the leakage and the unmasked value be only captured over multiple shares. The complexity of an attack therefore increases with the number of shares (see e.g. [5]).

However, masking defined this way is only secure under the assumption that the physical implementation is free from glitches and physical defects. This is not a realistic assumption in most cases [6, 7]. Such glitches appear when the output of a logic gate may change several times during a single clock cycle as a result of its inputs arriving at different times. As long as any sub-function depends on all input shares, the number of glitches that occur when executing this input may depend on all of its shares, and therefore on the unmasked input value. In case the number of glitches is related to some physical magnitude of the system (such as in CMOS circuits, where power consumption is significantly impacted by the occurrence of glitches), the protection gained by the sharing is lost.

1.2 Threshold Implementations

Threshold Implementations are masking schemes that aim at ensuring security against attacks of certain order, even in the presence of glitches and physical defects. TIs have a particular kind of dependence structure of the sub-functions on the input shares. By *dependence structure* we refer to the relationship between the set of the indices of the input shares $U_s = \{1, \dots, s\}$ and the set of indices of the sub-functions $U_r = \{1, \dots, r\}$. Following this dependence structure, an input share i and a sub-function j are related if and only if the j -th sub-function effectively depends on the i -th input share. Then, the sub-functions effectively depend on sets of the input shares which are subsets of U_s . As such, the dependence structure is represented by a set of subsets of U_s , with each set corresponding to a sub-function. And the elements of each set represent the indices of the input shares which the given sub-function may use.

In TIs, the dependence structure is such that any sub-function is not dependent on at least one input share for any input. Furthermore, for higher-order TIs, d sub-functions cannot depend on all the shares of any input. In this way, it is ensured that information about the execution of up to d sub-functions is independent of the unmasked input. This property is called *non-completeness*, and d is called the *order* of non-completeness of the TI. This value also defines the order of the attacks against which the implementation is secure. TIs were first proposed in [8] for 1st-order, and then extended to higher orders in [9].

Creating TIs to resist higher-order and multivariate attacks is not trivial. First, the security order d restricts the number of shares to $s \geq td + 1$. This greatly increases their implementation costs. Second, they do not guarantee a uniform split of the output shares [10]. This is critical for implementations that chain multiple masked functions, this being the case of most cryptographic algorithms (which are iterative). To solve this problem, the output of a TI-masked function must be *refreshed* [11]. Synchronization is also needed after each refresh operation [12]. Consequently, the number of output shares r has a direct impact on the randomness requirement and the number of registers.

The construction of TIs can be approached from the perspective of their dependence structures. Of particular relevance are dependence structures which, on top of being non-complete, have the property that all subsets of the input shares with cardinality t are subsets of at least one of their subsets. Such a dependence structure ensures that any monomial of degree up to t can be computed by one of its sub-functions, thus allowing for the construction of a TI for any Boolean function of algebraic degree up to t .

The problem of finding dependence structures to obtain TIs for Boolean functions of degree up to t is therefore equivalent to the problem of finding sets of subsets of U_s with the corresponding properties (non-completeness and coverage of all t -sets of U_s). This approach was formalized by Petrides in [13], where the author proposed the notion of *non-complete set coverings* (see Definition 1 in Section 2) and obtained a number of properties and specific instances of them. We identify particular cases of these objects with the notation $nc(s, t, d, r)$, but their general classes can be represented by the 3-tuple (s, t, d) . In the context of NCSCs, we will refer to s as the number of elements in the universe, to r as the number of sets in the covering (or its cardinality), to d as

the order of non-completeness (or the order of protection), and to t as the degree of coverage.

TIs are relevant for the masking of non-linear functions in cryptographic schemes, such as those that rely on S-Boxes (e.g. *SubBytes* in AES). Note that 4-bit S-Boxes have algebraic degree three, and 8-bit S-Boxes have algebraic degree up to seven. Third-degree NCSCs allow for direct secure computation of 4-bit S-Boxes in a single step. They also allow to compute 8-bit S-Boxes in two steps. For example, the inversion in $\mathbb{F}_{2^8}$, necessary for the masking of *SubBytes* in AES, can be decomposed into two cubic functions [14]. Fourth-degree NCSCs are also interesting, as fourth-degree functions have been proposed to speed up implementations of KECCACK by combining two quadratic operations in a single step [15].

For hardware implementation, the number of input shares and sub-functions affects the required resources (i.e. the physical space that the chip occupies) and efficiency (e.g. the throughput at which the circuit operates). As such, it is desirable that both s and r be as small as possible. An additional desirable property is that the number of input shares be equal, or at least approximate, to the number of sub-functions. This is so as the number of sub-functions determines the number of output shares and, given the iterative structure of many cryptographic algorithms, the outputs of a round are fed as inputs for the next round. We will refer to such NCSCs as *balanced*.

1.3 State of the Art

NCSCs were first defined and analyzed in [13], where a number of properties, together with a general strategy for the search of NCSCs of small cardinality is proposed and applied to the case of order $d = 2$ for degrees $t \in \{2, 3\}$.

For degree two and order two, a balanced NCSC made up of six elements and six sets is reported and shown to be optimal in [13]. To date, some practical coverings for degree three and order two have been found (see [13] and [16]), although their optimality in terms of s and r has not been established. The search for third-degree NCSCs was started in [13] for the case of second order, where a NCSC of eight elements with 18 sets is mentioned and one of nine elements with cardinality 13 is shown. For the same case of third degree and second order, [16] refers to NCSCs of eight elements with 17 sets, nine elements with 12 sets (both cases improve the coverings in [13]) and 10 elements with 10 sets. The latter one achieves a perfect balance between the number of elements and sets. For the case of degree $t = 3$ and order $d = 3$, NCSCs of 15, 16 and 17 elements with 46, 40 and 35 sets, respectively, are reported in [17]. In the literature, no references have been found for fourth-degree NCSCs. All these known NCSCs are summarized in Figure 1.

In addition to obtaining TIs for specific S-Boxes, previous works also tackled the problem of obtaining them for arbitrary S-Boxes. For instance, in [18] a method for obtaining a first-order TI for any S-Box of arbitrary degree is described. From the algorithmic perspective, [19] proposes methods for automating the process of seeking for a TI for some specific S-Box.

1.4 Our contributions

This work contributes to the study of NCSCs for efficient Threshold Implementations by finding smaller coverings and proving lower bounds on their size. We first present new theoretical results about NCSCs. We use these results to derive an algorithm for finding coverings. Using this method we obtain better NCSCs for the classes $(s, 3, 2)$, $(s, 3, 3)$, and $(s, 4, 2)$. As relevant results, we provide a lower bound for the number of sets in a covering in which any τ -set (where $\tau \in \{1 \dots t\}$) of U_s must be contained (Theorem 5). From this result, we derive lower bounds for the number of sets in a covering (Theorem 7 and Theorem 10). These bounds are used to evaluate the optimality of the new and the existing NCSCs.

In addition, we study the duality and shortening properties of NCSCs. We define a dual covering by transforming the sets and elements of the original into elements and sets of its dual, respectively (Lemma 11). We show that the degree and order of the original NCSC are swapped in the dual. Other theoretical results related to the existence of sub-coverings are presented in Lemma 12.

This work focuses afterwards on the analysis of NCSCs of the classes $(s, 3, 2)$, $(s, 3, 3)$, and $(s, 4, 2)$, which are particularly relevant in the context of TIs. For these classes of coverings we first obtain specific lower bounds on their cardinality (r). These bounds are derived from the theoretical results and complemented with *ad hoc* proofs of non-existence for specific cases. We then implement a computational search (Section 4) which leads to the discovery of specific NCSCs and consequently the proposal of an upper bound on r .

For the class $(s, 3, 2)$, Section 4.1 shows that the known NCSCs of 9 elements with 12 sets and 10 elements with 10 sets are optimal. As such, the best-balanced covering in this case is made up of ten elements and ten sets. One such covering is reported in Table A1. In addition, an NCSC of 11 elements and nine sets is obtained and also shown to be minimal.

For the class $(s, 3, 3)$, Section 4.2 reports NCSCs with smaller cardinality for the three known cases of $s \in \{15, 16, 17\}$. In addition, new coverings have been found for $s \in \{18, 19, 20\}$, including a balanced covering of 20 elements made up of 20 sets, which is reported in Table A2. Finally, it is shown that the minimal balanced covering must have at least 18 elements and 18 sets, and therefore a limited room for improvement remains.

For the class $(s, 4, 2)$, to the best of our knowledge, NCSCs have not been reported in the literature. In Section 4.3, we obtain NCSCs for $s \in \{10, \dots, 16\}$, including a balanced covering made up of 16 elements and 16 sets, which is reported in Table A3. Additionally, it is shown that the minimal balanced covering must have at least 15 elements and 15 sets, and therefore our best covering is either minimal or very close to minimal. All these results are summarized in Figure 1. Specific NCSCs of each of the aforementioned types are available at <https://github.com/oscar-fidalgo/ncscs>.

1.5 Organization

Section 2 provides a more formal description of Threshold Implementations and introduces the concept of non-complete set coverings and their main known properties.

	Petrides [13]		
$t = 2$ $d = 2$	$nc(5, 2, 2, 10)$ minimal $nc(6, 2, 2, 6)$ minimal		
$t = 3$ $d = 2$	$nc(7, 3, 2, 35)$ minimal $nc(8, 3, 2, 18)$ ← $nc(9, 3, 2, 13)$ ←	Grosso, Lara-Nino [16] $nc(8, 3, 2, 17)$ ← $nc(9, 3, 2, 12)$ ← $nc(10, 3, 2, 10)$ ←	This work minimal minimal $nc(11, 3, 2, 9)$ minimal
$t = 3$ $d = 3$		$nc(15, 3, 3, 46)$ ← $nc(16, 3, 3, 40)$ ← $nc(17, 3, 3, 35)$ ← Grosso, Lara-Nino [17]	$nc(15, 3, 3, 36)$ $nc(16, 3, 3, 30)$ $nc(17, 3, 3, 26)$ $nc(18, 3, 3, 22), nc(19, 3, 3, 20)$ $nc(20, 3, 3, 19), nc(20, 3, 3, 20)$ Best balanced: $s \in \{18, 19, 20\}$
$t = 4$ $d = 2$			$nc(10, 4, 2, 56), nc(11, 4, 2, 37)$ $nc(12, 4, 2, 28), nc(13, 4, 2, 22)$ $nc(14, 4, 2, 18), nc(15, 4, 2, 16)$ $nc(16, 4, 2, 15), nc(16, 4, 2, 16)$ Best balanced: $s \in \{15, 16\}$

Fig. 1 Summary of known NCSCs with $t \geq 3$ and the contribution of this work.

Afterwards, in [Section 3](#) we formulate theoretical results on the minimum number of repetitions of small subsets of U_s and provide lower bounds for the minimum number of sets in a NCSC. In addition, we define a way to derive dual NCSCs and also describe properties about the internal structure of NCSCs. [Section 4](#) describes a greedy search algorithm used to find NCSCs of small cardinality and the results obtained, together with the application of the theoretical results, in [Section 4.1](#), [Section 4.2](#) and [Section 4.3](#) for the classes $(s, 3, 2)$, $(s, 3, 3)$, and $(s, 4, 2)$, respectively. Finally, some open problems are discussed in [Section 5](#).

2 Preliminaries

We denote sets using capital letters, with $|S|$ denoting the cardinality of a set S and $\overline{S}$ its complement. We use the term τ -set to refer to subsets of U_s with cardinality τ . We use $C(n, k)$ for the number of ways to choose a sample of k elements from a set of n distinct objects where order does not matter and replacements are not allowed. Let $\mathbb{F}_2$ represent the binary field and $\mathbb{F}_2^n$ the n -dimensional vector space over $\mathbb{F}_2$. The elements of $\mathbb{F}_2^n$ can be regarded as bit arrays of length n . Let $x, y \in \mathbb{F}_2$, then xy and $x \oplus y$ represent the product (AND) and addition (XOR) in $\mathbb{F}_2$, respectively. Lastly, $\lceil z \rceil$ denotes the ceiling of the real number z , that is, the smallest integer greater than or equal to z .

2.1 Masking and Threshold Implementations

Each of the operations performed by cryptographic algorithms consists of the implementation of a Boolean vector function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$. To mask the execution of this function, the input of f , say x , is split into s shares, $x_1, \dots, x_s$, according to (1). A masking algorithm is said to be *uniform* if it chooses every possible mask with the same probability. As a result, uniform masking has the property that knowing up to $s - 1$ of the shares does not provide information about the masked input x .

In a masked implementation of f , the system implements a different (higher dimension) Boolean vector function $\tilde{f} : (\mathbb{F}_2^n)^s \rightarrow (\mathbb{F}_2^m)^r$ that (i) instead of taking a single n bit array x , takes the s shares $x_1, \dots, x_s$ as input, and (ii) instead of returning a single m bit array y , returns a sharing of y in r shares $y_1, \dots, y_r$, satisfying (2). This encapsulates the property of *correctness* of the masking. Each of the coordinate functions $\tilde{f}_i : (\mathbb{F}_2^n)^s \rightarrow \mathbb{F}_2^m$ such that $y_i = \tilde{f}_i(x_1, \dots, x_s)$ is called a *sub-function*.

A d th-order Threshold Implementation is a masking scheme in which the dependence structure of the sub-functions $\tilde{f}_i$ on the input shares x_j is such that any combination of up to d of the sub-functions does not make use of all the input shares. This property is called d th-order non-completeness. For example, consider the Boolean function $f(x, y) = xy$, which takes a 2-bit array as input and returns a single bit, for which a TI of first order is desired. The algebraic degree of this function is 2, then the number of required shares is $s = td + 1 = 3$. The inputs may be split as (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , where $x = x_1 \oplus x_2 \oplus x_3$ and $y = y_1 \oplus y_2 \oplus y_3$. Then, f can be expressed as a polynomial on the input shares as

$$\begin{aligned} f(x, y) &= xy \\ &= (x_1 \oplus x_2 \oplus x_3)(y_1 \oplus y_2 \oplus y_3) \\ &= x_1y_1 \oplus x_1y_2 \oplus x_1y_3 \oplus x_2y_1 \oplus x_2y_2 \oplus x_2y_3 \oplus x_3y_1 \oplus x_3y_2 \oplus x_3y_3. \end{aligned} \quad (3)$$

To split f into sub-functions, we need to spread the monomials in (3) among different sub-functions in such a way that no function depends on all three shares of (x, y) . For instance, reordering the monomials as convenient, we can define three sub-functions as follows:

$$\underbrace{x_1y_1 \oplus x_1y_2 \oplus x_2y_1 \oplus x_2y_2}_{\tilde{f}_1(x_1, x_2, y_1, y_2)} \oplus \underbrace{x_1y_3 \oplus x_3y_1 \oplus x_3y_3}_{\tilde{f}_2(x_1, x_3, y_1, y_3)} \oplus \underbrace{x_2y_3 \oplus x_3y_2}_{\tilde{f}_3(x_2, x_3, y_2, y_3)} \quad (4)$$

Note that $\tilde{f}_1$ does not depend on the third input share, $\tilde{f}_2$ does not depend on the second input share, and $\tilde{f}_3$ does not depend on the first input share. Every Boolean function can be expressed as a polynomial on its input shares [20, §2.2.1]. Therefore, to obtain a TI for a function f , we may first express it as a polynomial on the coordinates of the input shares and then spread the resulting monomials among the sub-functions $\tilde{f}_i$ with a d th-order of non-completeness dependence structure.

Note how, rather than the specific definition of the sub-functions, it is the dependence structure which is key in the derivation of a TI. A dependence structure in which every t -set of the input shares may be used by some of the sub-functions would ensure

the existence and guide the construction of a d th-order TI for any function of degree up to t . As an observation, it is worth mentioning that masking schemes can be seen as a particular kind of secret sharing schemes, and we point out how d -th order TIs are related to schemes whose access structure is Q_d (see [21, Chapter 6]).

2.2 Non-complete set coverings

NCSCs allow for the abstraction of the dependence structure of TIs. Therefore, they can be used as a guide for creating TIs of a given class (s, t, d) so long as they exist. In this subsection, we include their definition and basic properties. Those that will be used in the following sections are formulated as [Proposition 1](#) and [Theorem 2](#), and we refer to the work of Petrides [13] for their proofs.

Definition 1 ([13, Definition 2.1]) Given $s, t, d \in \mathbb{N}$, let $\mathcal{C}$ be a family of subsets of U_s . We say that $\mathcal{C}$ is a *non-complete covering* of U_s of degree t and order d if

1. every t -set of U_s is contained in at least one set of $\mathcal{C}$,
2. each set of $\mathcal{C}$ is of size not smaller than t , and
3. at least $d + 1$ sets of $\mathcal{C}$ are necessary to cover U_s .

In this case, we say that $\mathcal{C}$ is a $nc(s, t, d)$ covering for the (s, t, d) class, or a $nc(s, t, d, r)$ covering with $r = |\mathcal{C}|$ if we want to specify its cardinality.

From the previous section, it can be concluded that the existence of a $nc(s, t, d)$ covering ensures, through the first and third conditions of [Definition 1](#), that a TI can be obtained for any Boolean function of degree up to t . The second condition is not strictly necessary, since the first one already guarantees that every monomial of degree up to t can be computed by at least one of the functions. This is so because any subset of size smaller than t is included in some t -set, which in turn is included in some set of the covering. This second condition is therefore aimed at avoiding the inclusion of redundant small sets. However, note that a similar condition is not introduced in the definition to avoid redundant sets of larger sizes. For example, according to this definition, a $nc(s, 3, d)$ covering can include the sets $\{1, 2, 3, 4\}$ and $\{1, 2, 3, 4, 5, 6\}$, although the first is redundant and might be removed. As such, a trivial increase in the number of sets is possible according to this definition. In any case, the original definition in [13] will be used in this work.

The following proposition (which, although already known in the context of TIs, was first formulated in the language of NCSCs in [13]) imposes a lower bound on the number of elements that are necessary for an NCSC of degree t and order d to exist.

Proposition 1 ([13, Proposition 2.5]). *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering. Then, $s \geq td + 1$.*

If $s \geq td + 1$, then the set of all t -sets of U_s is a $nc(s, t, d)$ covering. Although not very practical (since it is made up of $r = C(s, t)$ sets), this covering represents an upper bound on the minimal number of sets in an NCSC when t and d are fixed. Moreover, if $s = td + 1$, this covering is in fact minimal. Furthermore, increasing s can

lead to finding coverings with a smaller or equal number of sets. This follows from the fact that for any $nc(s, t, d, r)$ covering $\mathcal{C}$, there exists a $nc(s + 1, t, d, r)$ covering. It is enough to consider $\mathcal{C}' \subseteq \mathcal{C}$ covering all $(t - 1)$ -sets of U_s , and add the element $s + 1$ to each subset in $\mathcal{C}'$. (See [13] for further details and proofs of all these properties.)

Finally, another basic result that will be used in this work relates the maximum size that a set in a covering may have with the degree t and order d , according to the following theorem.

Theorem 2 ([13, Corollary 2.4]). *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering. Then, every $S \in \mathcal{C}$ is of size at most $s - (d - 1)t - 1$.*

3 Properties of non-complete set coverings

In this section, we formulate new properties of NCSCs. The results in Section 3.1 refer to the coverage of τ -sets of U_s , for τ smaller than the degree of the covering t . From this result, we derive lower bounds for the number of sets in a covering (Theorem 7 and Theorem 10). In Section 3.2 and Section 3.3, we present theoretical results on duality and shortening properties.

3.1 Lower bounds for the cardinality of a covering

Let $\mathcal{C}$ be a $nc(s, t, d, r)$ covering. From Definition 1, all the different $C(s, t)$ t -sets of U_s must be contained in the sets of $\mathcal{C}$. Note also that the cardinality of any set in $\mathcal{C}$ is at most $s - t(d - 1) - 1$, according to Theorem 2. Therefore, any set in $\mathcal{C}$ contains up to $C(s - t(d - 1) - 1, t)$ t -sets. From this, we can conclude that

$$r \geq \frac{C(s, t)}{C(s - t(d - 1) - 1, t)}. \quad (5)$$

Tighter lower bounds on the cardinality of the covering can be obtained if, instead of counting the minimum number of t -sets that must be contained in it, we shift the focus towards the number of its τ -sets, where $\tau \in \{1 \dots t\}$. We can first give a lower bound for the number of sets in the covering in which every $(t - 1)$ -set must be contained.

Lemma 3. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering. Then, every $(t - 1)$ -set of U_s is contained in at least $d + 1$ sets of $\mathcal{C}$.*

Proof Let X be a $(t - 1)$ -set of U_s . For the sake of contradiction, suppose that X is contained in exactly k sets of $\mathcal{C}$, with $k \leq d$. Let $Y_1, \dots, Y_k \in \mathcal{C}$ be such sets. Observe that, for every $i \in U_s \setminus X$, the set $X \cup \{i\}$ must be contained in at least one set of $\mathcal{C}$. Hence, $X \cup \{i\} \subseteq Y_j$ for some $1 \leq j \leq k$. Therefore, $Y_1 \cup \dots \cup Y_k = U_s$, and so the order of $\mathcal{C}$ is smaller than k , resulting in a contradiction. $\square$

Additionally, the following lemma provides a minimum number of repetitions for sets of some size smaller than t based on a minimum number of repetitions of the sets having one more element.

Lemma 4. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering whose largest set is of size m such that for some $\tau \in \{1 \dots t-1\}$, every $(\tau+1)$ -set of U_s is contained in at least $\sigma_{\tau+1}$ sets of $\mathcal{C}$. Then, every τ -set of U_s is contained in at least*

$$\left\lceil \frac{\sigma_{\tau+1}(s-\tau)}{m-\tau} \right\rceil$$

sets of $\mathcal{C}$.

Proof Let $\tau \in \{1 \dots t-1\}$, let X be a τ -set of U_s . Let $S_1, \dots, S_k \in \mathcal{C}$ be the sets of the cover containing X . Define $A_X = \{X \cup \{y\} : y \in U_s \setminus X\}$. That is, A_X is the set of all $(\tau+1)$ -sets of U_s containing X . Notice that $|A_X| = (s-\tau)$. Since every element in A_X is contained in at least $\sigma_{\tau+1}$ sets of $\mathcal{C}$, the sets of $\mathcal{C}$ contain at least $\sigma_{\tau+1} |A_X|$ elements of A_X (counting all repetitions). That is,

$$\sum_{j=1}^k |\{Y \in A_X : Y \subseteq S_j\}| \geq \sigma_{\tau+1} |A_X| = \sigma_{\tau+1}(s-\tau).$$

Now we give an upper bound for the left-hand side of the previous inequality. Each S_i contains $|S_i| - \tau$ elements of A_X . Namely, all $(\tau+1)$ -sets of the form $X \cup \{y\}$ with $y \in S_i \setminus X$. Taking into account that $|S_i| \leq m$, we have that

$$\sum_{j=1}^k |\{Y \in A_X : Y \subseteq S_j\}| \leq \sum_{i=1}^k (|S_i| - \tau) \leq k(m - \tau),$$

which implies that X is in $k \geq \sigma_{\tau+1}(s-\tau)/(m-\tau)$ subsets of $\mathcal{C}$. $\square$

Based on these two lemmas, we can elaborate as follows. Given any $nc(s, t, d)$ covering whose largest set has cardinality m , from [Lemma 3](#) we know that each $(t-1)$ -set of U_s is contained in at least $\sigma_{t-1} = d+1$ sets of the covering. Then, from [Lemma 4](#), we have that each $(t-2)$ -set of U_s is contained in at least

$$\sigma_{t-2} = \left\lceil \frac{\sigma_{t-1}(s-t+2)}{m-t+2} \right\rceil$$

sets of $\mathcal{C}$. We can continue iterating until we reach σ_1 , the minimum number of sets of $\mathcal{C}$ that contains every element of U_s . This is compiled in the following theorem.

Theorem 5. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering whose largest set is of size m . Then, for every $\tau \in \{1, \dots, t\}$, every τ -set of U_s is contained in at least σ_τ sets of $\mathcal{C}$, where σ_τ is defined as*

$$\sigma_\tau = \begin{cases} 1 & \tau = t \\ d+1 & \tau = t-1 \\ \left\lceil \frac{\sigma_{\tau+1}(s-\tau)}{m-\tau} \right\rceil & 1 \leq \tau < t-1 \end{cases} \quad (6)$$

From now on, σ_τ will be used to denote the lower bound on the number of sets in the covering that contain any τ -set of U_s according to Equation (6). Once a minimum number of necessary repetitions of τ -sets has been established for $\tau \in \{1, \dots, t\}$, we can now focus on the number and size of the sets of the covering that would be needed to include all of them. For that, given any covering $\mathcal{C}$, let r_k denote the number of its subsets with cardinality k . That is,

$$r_k \stackrel{\text{def.}}{=} |\{S \in \mathcal{C} : |S| = k\}|. \quad (7)$$

As such, r_k is defined for $t \leq k \leq s - t(d-1) - 1$, where the lower bound is given by Definition 1, and the upper bound is given by Theorem 2. The total number of sets in the covering is $|\mathcal{C}| = \sum_k r_k$. The following proposition relates the number of τ -sets that a covering must allocate with the number and sizes of its sets.

Proposition 6. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering. Then,*

$$\sum_{k=t}^{s-t(d-1)-1} r_k C(k, \tau) \geq \sigma_\tau C(s, \tau) \quad \forall \tau \in \{1, \dots, t\} \quad (8)$$

Proof Let $\tau \in \{1, \dots, t\}$. We prove the result by using a double-counting argument. We count the total number of τ -sets contained in the sets of $\mathcal{C}$ (with repetitions), that is

$$\sum_{|X|=\tau} |\{Y \in \mathcal{C} : X \subseteq Y\}| = \sum_{Y \in \mathcal{C}} |\{X \subseteq Y : |X| = \tau\}|.$$

Observe that, in the right-hand side, for every $Y \in \mathcal{C}$ with $|Y| = k$ we have that $|\{X \subseteq Y : |X| = \tau\}| = C(k, \tau)$. Since there are r_k sets of size k in $\mathcal{C}$, we obtain the right-hand side is equal to

$$\sum_{k=t}^{s-t(d-1)-1} r_k C(k, \tau).$$

Now, according to Theorem 5, each τ -set is contained in at least σ_τ sets of the covering. Since the number of τ -sets is $C(s, \tau)$, we have that the left-hand term is at least $\sigma_\tau C(s, \tau)$. $\square$

Based on the condition in Proposition 6, a lower bound can be obtained for the number of sets in the covering as a function of the size of its largest set.

Theorem 7. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering such that its largest set is of size m . Then,*

$$|\mathcal{C}| \geq \max_{1 \leq \tau \leq t} \left\{ \sigma_\tau \frac{C(s, \tau)}{C(m, \tau)} \right\} \quad (9)$$

Proof We follow the counting argument of the proof of Proposition 6.

Let $\tau \in \{1, \dots, t\}$. Since every set in $\mathcal{C}$ is of size at most m , the total number of τ -sets contained in the sets of $\mathcal{C}$ (with repetitions) is at most $|\mathcal{C}|C(m, \tau)$. Then, by [Proposition 6](#), we have that

$$|\mathcal{C}|C(m, \tau) \geq \sum_{k=t}^{s-t(d-1)-1} r_k C(k, \tau) \geq \sigma_\tau C(s, \tau),$$

which implies that $|\mathcal{C}| \geq \sigma_\tau C(s, \tau)/C(m, \tau)$. Note that this bound must hold for every value of τ . $\square$

At first glance, having as large sets as possible in the covering seems beneficial for obtaining NCSC of small cardinality. This seems intuitive as increasing the maximum set size m may reduce the lower bound on the number of necessary repetitions of τ -sets, σ_τ for $1 \leq \tau < t-1$, according to [Theorem 5](#). This may also reduce the lower bound obtained according to [Theorem 7](#). And this would always be the case in the absence of any other restrictions. However, for cases where $d \geq 3$ there is a trade-off to be considered: large sets can also lead to increase the cardinality of the NCSC as shown below.

First, note that for any $d \geq 2$, given any $d-1$ sets of the covering, the number of elements of U_s that are not included in any of them cannot be too small. This is schematically illustrated in [Figure 2](#) and formally stated in [Proposition 8](#).

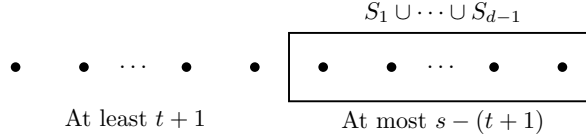


Fig. 2 Minimum size of $\overline{S_1 \cup \dots \cup S_{d-1}}$.

Proposition 8. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering with $d \geq 2$ and let $S_1, \dots, S_{d-1} \in \mathcal{C}$. Then, there exist $t+1$ elements $u_1, \dots, u_{t+1} \in U_s$ satisfying that $u_i \notin S_j$ for every $1 \leq i \leq t+1$ and $1 \leq j \leq d-1$.*

Proof Given any $d-1$ sets of $\mathcal{C}$, $S_1, \dots, S_{d-1}$, consider the set of the elements of U_s that are not contained in any of them. That is, $X = \overline{S_1 \cup \dots \cup S_{d-1}}$. Note that X is not contained in any other set $S \in \mathcal{C}$, as otherwise the d sets $S, S_1, \dots, S_{d-1}$ would cover U_s .

Then, if $|X| = t$, X would be a t -set of U_s that is not contained in any of the sets of $\mathcal{C}$. Furthermore, if $|X| < t$, all the t -sets containing X would not be contained in any set of the covering. Both cases contradict [Definition 1](#). Therefore, $|X| \geq t+1$. $\square$

Remark 1 For the case where $d = 2$, [Proposition 8](#) leads to the need that the complement of any set of the covering be of size at least $t+1$. Or equivalently, the size of any set of the covering is at most $s - t - 1$. This is exactly the same result as in [Theorem 2](#) for $d = 2$. As such, [Proposition 8](#) does not represent any additional constraint to those considered so far for the case where $d = 2$. However, for the cases where $d \geq 3$, this result can be used to obtain further restrictions that such coverings must fulfill, which in turn can be used to obtain tighter lower bounds for the number of sets of the covering.

Consider a $nc(s, t, d)$ covering $\mathcal{C}$ with $d \geq 3$. Consider also any $d - 2$ sets $S_1, \dots, S_{d-2} \in \mathcal{C}$ whose union is of size m , and hence the number of elements of U_s not included in any of these sets is $|\overline{S_1 \cup \dots \cup S_{d-2}}| = s - m$. Now, let r'_k denote the number of sets of $\mathcal{C}$ containing exactly k elements of $\overline{S_1 \cup \dots \cup S_{d-2}}$. That is,

$$r'_k \stackrel{\text{def.}}{=} \{S \in \mathcal{C} : |(\overline{S_1 \cup \dots \cup S_{d-2}}) \cap S| = k\}.$$

Any $S \in \mathcal{C} \setminus \{S_1, \dots, S_{d-2}\}$ may contain a number of these $s - m$ elements in $\overline{S_1 \cup \dots \cup S_{d-2}}$ not larger than $(s - m) - (t + 1)$ as a result of [Proposition 8](#). This is illustrated in [Figure 3](#). As such, r'_k is defined for $0 \leq k \leq (s - m) - (t + 1)$.

Note that, as m grows, the number of elements in $\overline{S_1 \cup \dots \cup S_{d-2}}$ that can be contained in the sets of the covering decreases. This has an impact on the number of sets in the covering necessary to cover the t -sets of U_s contained in $\overline{S_1 \cup \dots \cup S_{d-2}}$.

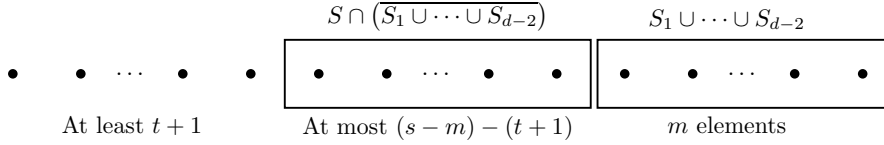


Fig. 3 Maximum size of $S \cap (\overline{S_1 \cup \dots \cup S_{d-2}})$.

Proposition 9. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering with $d \geq 3$ satisfying that there exist $d - 2$ sets $S_1, \dots, S_{d-2} \in \mathcal{C}$ with $|S_1 \cup \dots \cup S_{d-2}| = m$. Then,*

$$\sum_{k=\tau}^{(s-m)-(t+1)} r'_k C(k, \tau) \geq \sigma_\tau C(s - m, \tau) \quad \forall \tau \in \{1, \dots, t\}$$

Proof Let $\tau \in \{1, \dots, t\}$. Define $Y = \overline{S_1 \cup \dots \cup S_{d-2}}$. Any $S \in \mathcal{C}$ such that $|S \cap Y| = k$ with $k \geq \tau$ contains exactly $C(k, \tau)$ τ -sets of Y . If $k < \tau$, no τ -set of Y is contained in S . Recall that for any $k \in \{0, \dots, (s - m) - (t + 1)\}$, there are r'_k such sets of $\mathcal{C}$ with $|S \cap Y| = k$. They contain $r'_k C(k, \tau)$ τ -sets of Y in total (with repetitions). Summing across all possible sizes of the intersection, we obtain the total number of τ -sets of Y contained in the covering is

$$\sum_{k=\tau}^{(s-m)-(t+1)} r'_k C(k, \tau). \tag{10}$$

Notice that $|Y| = s - m$. As such, there are $C(s - m, \tau)$ τ -sets contained in Y . Per [Theorem 5](#), each of them must be contained in at least σ_τ sets of $\mathcal{C}$. Therefore, at least $\sigma_\tau C(s - m, \tau)$ τ -sets in Y must be contained in the sets of the covering. Hence, $\sigma_\tau C(s - m, \tau)$ is a lower bound of [Equation \(10\)](#). $\square$

Now, considering that the maximum number of τ -sets in the covering is attained in the case where the intersections with $\overline{S_1 \cup \dots \cup S_{d-2}}$ are as large as possible, another lower bound for the number of sets in the covering may be obtained.

Theorem 10. Let $\mathcal{C}$ be a $nc(s, t, d)$ covering where $d \geq 3$ such that there exist $S_1, \dots, S_{d-2} \in \mathcal{C}$ with $|S_1 \cup \dots \cup S_{d-2}| = m$. Then,

$$|\mathcal{C}| \geq \max_{1 \leq \tau \leq t} \left\{ \sigma_\tau \frac{C(s-m, \tau)}{C(s-m-t-1, \tau)} \right\} + d - 2 \quad (11)$$

Proof Define $Y = \overline{S_1 \cup \dots \cup S_{d-2}}$. Let $\tau \in \{1, \dots, t\}$. Note that the τ -sets of Y are contained in sets of $\mathcal{C}$ other than $S_1, \dots, S_{d-2}$. The number of these other sets is $|\mathcal{C}| - (d - 2)$.

As discussed above (see Figure 3), the maximum size of the intersection of these other sets with Y is $(s - m) - (t + 1)$. A set with such maximal intersection would contain $C((s - m) - (t + 1), \tau)$ τ -sets of Y . The number of τ -sets of Y contained in $\mathcal{C}$ would therefore be maximal if all the $|\mathcal{C}| - (d - 2)$ sets other than $S_1, \dots, S_{d-2}$ have an intersection with Y of size $(s - m) - (t + 1)$. In this case, they would contain $(|\mathcal{C}| - (d - 2)) C((s - m) - (t + 1), \tau)$ τ -sets of Y . This represents an upper bound for the number of τ -sets of Y in Equation (10), leading to

$$(|\mathcal{C}| - (d - 2)) C((s - m) - (t + 1), \tau) \geq \sum_{k=\tau}^{(s-m)-(t+1)} r'_k C(k, \tau).$$

From Proposition 9, the right hand side of the previous inequality must be larger than $\sigma_\tau C(s - m, \tau)$. Therefore,

$$(|\mathcal{C}| - (d - 2)) C((s - m) - (t + 1), \tau) \geq \sigma_\tau C(s - m, \tau),$$

obtaining that

$$|\mathcal{C}| \geq \frac{\sigma_\tau C(s - m, \tau)}{C((s - m) - (t + 1), \tau)} + (d - 2).$$

□

Remark 2 The dependence of the lower bound of Theorem 10 on m is not monotonic. For instance, for the case of a $nc(20, 3, 3)$ covering, m can range from 3 to 13 (according to Theorem 2). For $m = 3$, the value of this bound is 51. From $m = 3$ to $m = 10$ the bound monotonically decreases to a value of 16. From $m = 11$ to $m = 13$ it monotonically increases from 16 to 36.

3.2 Dual coverings

Given any NCSC, a dual covering may be obtained. The sets in the covering are the elements in the dual; the elements in the covering are the sets in the dual; and the inclusion relationship in the dual is the opposite of that in the covering.

Let $\mathcal{C} = \{S_1, \dots, S_r\}$ be a $nc(s, t, d, r)$ covering. For any $i \in U_s$, consider the set $S_i^* \subset U_r$ defined based on $\mathcal{C}$ as follows: for any $j \in U_r$, we have $j \in S_i^*$ if and only if $i \notin S_j$. That is,

$$S_i^* \stackrel{\text{def.}}{=} \{j \in U_r : i \notin S_j\} \subset U_r$$

Let $\mathcal{C}^*$ be the set of sets of U_r made up of all sets S_i^* , for $i \in U_s$, whose size is larger than d . That is,

$$\mathcal{C}^* \stackrel{\text{def.}}{=} \{S_i^* : |S_i^*| \geq d\}. \quad (12)$$

The construction of each of the sets S_i^* and $\mathcal{C}^*$ is illustrated in Table 1. The check marks in the table represent a $nc(7, 2, 2, 6)$ covering. Each column corresponds to an element of U_7 , each row corresponds to a set in the covering, a check mark indicates that the element of the column is included in the set of the row, and, conversely, a square indicates that the element in the column is not in the set of the row. For the construction of the S_i^* and $\mathcal{C}^*$, the rows are viewed as the elements of U_6 and the columns as the sets S_i^* . The squares indicate that the element of the row is in the set of the column, and, conversely, check marks indicate that the element of the row is not in the set of the column. In this particular case, all the sets S_i^* are made of at least two (d) elements, and $\mathcal{C}^*$ is made of the seven (s) S_i^* .

Table 1 Example: a $nc(7, 2, 2, 6)$ covering and its dual $nc(6, 2, 2, 7)$ covering

	1	2	3	4	5	6	7	
S_1	✓	✓	✓	✓	□	□	□	1^*
S_2	□	□	✓	✓	✓	✓	□	2^*
S_3	✓	□	✓	□	✓	□	✓	3^*
S_4	✓	□	□	✓	□	✓	✓	4^*
S_5	□	✓	□	□	□	✓	✓	5^*
S_6	□	✓	□	□	✓	□	✓	6^*
	S_1^*	S_2^*	S_3^*	S_4^*	S_5^*	S_6^*	S_7^*	

The following proposition shows that $\mathcal{C}^*$ is indeed a $nc(s, d, t)$ covering.

Lemma 11. *Let $\mathcal{C}$ be a $nc(s, t, d, r)$ covering. Then $\mathcal{C}^*$ is a $nc(r, d, t, s')$ covering for some $s' \leq s$.*

Proof We prove that $\mathcal{C}^*$ meets the conditions 1, 2, and 3 of a $nc(r, d, t, s')$ covering from Definition 1. By construction, every set in $\mathcal{C}^*$ is made up of at least d elements of U_r , thus condition 2 is fulfilled.

Suppose that condition 1 is not satisfied. That is, there exists some $X \subseteq U_r$ with $|X| \leq d$ that is not included in any set of $\mathcal{C}^*$. It implies that every $i \in U_s$ is contained in the union of S_j 's with $j \in X$, contradicting the d -th order non-completeness of $\mathcal{C}$ (condition 3).

Now suppose that condition 3 is not satisfied. Suppose that t sets of $\mathcal{C}^*$ cover U_r . Then there exist t elements of U_s not included in any set of $\mathcal{C}$, contradicting the coverage of all t -sets in $\mathcal{C}$ (condition 1).

Note that the number of sets in the dual covering is s if all S_i^* were of size at least d , and smaller otherwise. Hence, $|\mathcal{C}^*| \leq s$. $\square$

It can be seen from the previous proof that the non-completeness of the original covering leads to the t -sets coverage in the dual, and conversely, the t -sets coverage of the original leads to the non-completeness in the dual. If these were the only two properties defining non-complete set coverings (i.e. if condition 2, which as discussed in Section 2 is not strictly necessary, was removed from Definition 1), the dual operation might simply be defined as $\{S_i^*\}$. In that case, it would be an involution over NCSCs (i.e. $(\mathcal{C}^*)^* = \mathcal{C}$), which is a property that other definitions of dual mathematical objects

typically have. That said, as mentioned in [Section 2](#), we will maintain the definition of NCSCs from [\[13\]](#) and the definition of the $*$ operation given in [Equation \(12\)](#).

Remark 3 An alternative way to obtain an operation which is an involution over NCSCs would be to add the dual property of condition 2 as an additional condition of [Definition 1](#). Namely, adding the following fourth condition to the definition of NCSCs:

4. Every element of U_s is contained in less than $s - d$ sets of the covering.

Note that this condition translates in the dual to condition 2, as would imply that in the dual no set has less elements than the degree of the dual covering (d). With this condition, every set in the dual would be of size larger or equal to d , and therefore the condition that $|S_i^*| \geq d$ may again be omitted from the definition of the $*$ operation. In fact, just like condition 2 aims at avoiding redundant sets not necessary for the t -sets coverage, such condition 4 would avoid the inclusion of redundant elements not necessary for the non-completeness property. Indeed, if a covering would have an element contained in $s - d$ elements or more, that element might be removed from all the sets in which it appears, and the new sets would still necessarily comply with non-completeness.

3.3 Sub-coverings

This subsection includes some results related to the internal sub-coverings structure that an NCSC must have. Although not directly used in our work, sub-coverings may be considered of general usefulness and might be further developed to assist in the assessment of whether NCSCs for certain parameters exist and their construction.

Definition 2 Let $\mathcal{C}$ be a $nc(s, t, d)$ covering and x any element of U_s . The set of sets of $\mathcal{C}$ that contains x will be denoted $\mathcal{C}_x$. That is,

$$\mathcal{C}_x \stackrel{\text{def.}}{=} \{S \in \mathcal{C} : x \in S\}.$$

The operation of removing x from all the sets in $\mathcal{C}$ where it is contained and then renaming the set $U_s \setminus \{x\}$ through any bijection $U_s \setminus \{x\} \rightarrow U_{s-1}$ will be denoted $\mathcal{C} - x$.

Lemma 12. *Let $\mathcal{C}$ be a $nc(s, t, d)$ covering, and consider any element $x \in U_s$. Then $\mathcal{C}_x$ is a $nc(s, t - 1, d)$ covering and $\mathcal{C}_x - x$ is a $nc(s - 1, t - 1, d)$ covering.*

Proof Every t -set of the form $X \cup \{x\}$, where X is any $(t - 1)$ -set of $U_s \setminus \{x\}$, is included in some set of $\mathcal{C}_x$. Therefore, all $(t - 1)$ -sets (either containing x or otherwise) are still included in $\mathcal{C}_x$. Moreover, since any set in $\mathcal{C}$ is of size not smaller than t , any element in $\mathcal{C}_x$ is of size not smaller than t . Finally, if there were d or fewer sets of $\mathcal{C}_x$ covering U_s , $\mathcal{C}$ would not be a $nc(s, t, d)$ covering.

Observe that the family obtained by removing x from $\mathcal{C}_x$, that is $\mathcal{C}_x - x$, still fulfills the conditions of NCSCs for the remaining elements. $\square$

[Lemma 12](#) is related to the necessary internal structure that any covering must have. More specifically, a $nc(s, t, d, r)$ covering can be seen as the combination of s $nc(s, t - 1, d)$ sub-coverings, each of which has an element of U_s in all its sets, such that

they share common sets to an extent that only r different sets are used in all these s coverings. Further work on the conditions under which this combination is possible with r sets might be useful for the purpose of discerning whether some $nc(s, t, d, r)$ exists, and eventually to derive some constructive method.

4 Computational search

A randomized greedy algorithm has been implemented to search for NCSCs. Even if this approach does not provide any guarantee that the obtained NCSCs are minimal, it allows obtaining some good coverings that serve to determine the upper bounds on the cardinality of NCSCs of a given class. Previous works on automating the search of TIs for a given S-Box also considered randomized greedy algorithms (see [19]). Our approach is different in two ways. Firstly, our focus is on finding NCSCs, rather than directly looking for TIs for specific S-Boxes. We note that for higher-order TIs additional gadgets are required to ensure uniformity, and thus the assignment of the monomials in the shared function is not relevant. Secondly, instead of subsequently allocating the monomials of the function to be implemented to the different sub-functions (similar to allocating t -sets to the sets of the covering in the language of NCSCs), our approach is to subsequently add sets to the covering, keeping track of all the t -sets that each such set covers. For this, we leveraged the theoretical results from the previous section to reduce the search space (in particular, by reducing the sets which are candidate for addition in each step on the basis of [Proposition 8](#) and [Theorem 10](#)).

Given s , t , d , and m , the latter being the maximal size of the sets in the covering, the approach to obtain a $nc(s, t, d)$ covering is as follows. The first set added to the covering is $\{1, 2, \dots, m\}$. Note that there is no loss of generality, as for any covering containing a set S of size m , there exists a permutation of the elements of U_s such that the set S becomes the set $\{1, 2, \dots, m\}$. Then, additional sets are subsequently added by randomly choosing among those fulfilling the non-completeness condition that maximize the number of not yet covered t -sets of U_s . This process is repeated until all the t -sets of U_s are covered. It might be conjectured that in the selection of the set to be added, choosing one which is not among the ones that maximize the additional t -sets in one step might eventually lead to the possibility of more efficiently covering the remainder of the t -sets. To account (to some extent) for that possibility, the algorithm was prepared to choose the next set randomly among those that would add the maximum possible number of t -sets (x), or a close number. This was done by considering a parameter $k \in (0, 1]$ such that the choice is made over those sets that would add at least kx t -sets. The search consists in iterating the previous process many times and retrieving the obtained covering with the minimum cardinality. Note that it is not explicitly imposed that all the added sets be of a large size. However, the fact that the selection is made over those that maximize the number of additional t -sets covered implicitly leads to selecting large sets, with the restriction that the maximum set size is m .

This algorithm has been implemented for NCSCs of orders two and three. For this purpose, a NCSC, denoted $\mathcal{C}$, is represented by (i) a list of the sets

already added to the covering, denoted $\mathcal{C}.\text{sets}$; (ii) a list of the t -sets not yet covered, denoted $\mathcal{C}.\text{t_sets_pending}$; and (iii) a list of candidate sets for addition denoted $\mathcal{C}.\text{set_options}$. Each set option $S \in \mathcal{C}.\text{set_options}$ carries the information of the number of additional t -sets that its inclusion would provide, denoted $S.\text{n_t_sets_added}$, and its size, denoted $S.\text{set_size}$. The creation of an empty covering $\mathcal{C}$, denoted $\mathcal{C}.\text{create}(s, t, d, m)$ consists on the allocation of these three lists with $\mathcal{C}.\text{sets}$ being empty, $\mathcal{C}.\text{t_sets_pending}$ initialized with all t -sets, and $\mathcal{C}.\text{set_options}$ with all subsets $X \subset U_s$ with $t \leq |X| \leq m$. The operation of adding a set S to the covering is denoted $\mathcal{C}.\text{add}(S)$. On top of adding S to $\mathcal{C}.\text{sets}$, it updates both $\mathcal{C}.\text{t_sets_pending}$ (by dropping the t -sets contained in S) and $\mathcal{C}.\text{set_options}$ (by dropping those whose addition would lead to violating the non-completeness¹, updating $S.\text{n_t_sets_added}$ for the remaining ones and dropping those that do not contain any t -set pending). This allows for quick greedy selection of the next set to add. [Algorithm 1](#) formalizes the aforementioned procedure, where $\mathcal{U}(X)$ denotes the uniform selection of an element of a set X .

Algorithm 1 Randomized greedy search

Require: Number of elements s , degree t , order d , maximum set size m (takes $s - t(d - 1) - 1$ by default), and a tolerance parameter $k \in (0, 1]$.

Ensure: A $nc(s, t, d)$ covering with maximum set size m .

```

1:  $\mathcal{C}.\text{create}(s, t, d, m)$ 
2:  $\mathcal{C}.\text{add}(\{1, 2, \dots, m\})$ 
3: while  $\mathcal{C}.\text{t\_sets\_pending}$  not empty do
4:    $x \leftarrow \max \{S.\text{n\_t\_sets\_added} : S \in \mathcal{C}.\text{set\_options}\}$ 
5:    $O \leftarrow \{S \in \mathcal{C}.\text{set\_options} : S.\text{n\_t\_sets\_added} \geq kx\}$ 
6:    $n \leftarrow \min \{S.\text{set\_size} : S \in O\}$ 
7:    $O \leftarrow \{S \in O : S.\text{set\_size} = n\}$ 
8:    $S \leftarrow \mathcal{U}(O)$ 
9:    $\mathcal{C}.\text{add}(S)$ 
10: end while
```

In [13, §2.4.1], a general strategy for the search of small coverings of degree 2 is defined. Despite the somewhat different formulations, for the particular case where $d = 2$ our greedy search algorithm resembles that strategy to a large extent. Both procedures start from a set of maximal size. Afterward, additional sets are subsequently added until all t -sets are covered in both cases, although with a slightly different logic. The general strategy selects the t -sets not yet covered from the complement of previously added sets and completes them with additional elements until they reach the maximum size. In our case, we directly chose whole additional sets among the best candidates, including some tolerance for sub-optimality in the selected set. Some of the minimal coverings obtained resulted from relying on this tolerance.

¹For the case $d = 2$ this is done by dropping all set options that, together with the set being added, would cover U_s . For the case $d = 3$ the process loops all the set that were already in the covering. For each such set, S , the process drops all set options that, together with S and with the set being added, would fully cover U_s .

This search was implemented for on the classes $(s, 3, 2)$, $(s, 3, 3)$ and $(s, 4, 2)$. The sizes of the specific coverings of minimal size that were obtained are reported in the following sections as upper bounds for the cardinality of minimal coverings. In addition, Table A1, Table A2 and Table A3 in Appendix A show the balanced coverings of minimal size obtained for each case, namely $nc(10, 3, 2, 10)$, $nc(20, 3, 3, 20)$ and $nc(16, 4, 2, 16)$. An implementation of the search algorithm and NCSCs of the minimal size obtained for each type are available at <https://github.com/oscar-fidalgo/ncscs>.

By default, the algorithm runs a predefined number of repetitions. Although these were sufficient for most cases, we followed an interactive approach extending the number of repetitions on a case by case basis. As a result of the randomized nature of the algorithm, it cannot be discarded that running it for longer would lead to smaller coverings than those obtained, eventually converging to the lower bounds derived for each case.

4.1 Class $(s, 3, 2)$ coverings

The application of Theorem 5 to this particular case yields the following minimum number of repetitions of τ -sets in a $nc(s, 3, 2)$ covering whose largest set is of size m , for $1 \leq \tau \leq 3$:

$$\begin{aligned}\sigma_3^{s,3,2} &= 1 \\ \sigma_2^{s,3,2} &= 3 \\ \sigma_1^{s,3,2} &= \left\lceil \frac{3(s-1)}{m-1} \right\rceil\end{aligned}\tag{13}$$

From Theorem 7 we get the following lower bound on the cardinality of the covering depending on the size of its largest set:

$$|C| \geq \max \left\{ \left\lceil \frac{3(s-1)}{m-1} \right\rceil, \frac{s}{m}, \frac{3C(s,2)}{C(m,2)}, \frac{C(s,3)}{C(m,3)} \right\}.\tag{14}$$

According to Remark 1, the bound of Theorem 10 does not apply in this case ($d = 2$).

The second and third columns of Table 2 show the minimum number of repetitions and the lower bounds obtained according to Theorem 7 for different values of s . These lower bounds are obtained where m is of maximum size $m = s - t(d-1) - 1$, that is $m = s - 4$ in this case. The blue line in Figure 4 displays these bounds.

From the computational search described in the previous section, the NCSCs $nc(8, 3, 2, 17)$, $nc(9, 3, 2, 12)$, $nc(10, 3, 2, 10)$ and $nc(11, 3, 2, 9)$ were obtained. These coverings contributed to define the upper bounds on r , which are included in the last column of Table 2 and graphically in the red line in Figure 4. As mentioned above, the existence of coverings such as the first three was already reported in [16].

From the combination of the upper and lower bounds, we conclude that the best possible balanced covering of degree $t = 3$ and order $d = 2$ is made of 10 elements, since a covering of nine elements would have at least 11 sets. As such, the $nc(10, 3, 2, 10)$ covering used in [16] is the optimal balanced covering for this degree and order. One such covering is shown in Table A1.

Table 2 Results of the search of the $(s, 3, 2)$ class of coverings.

s	σ_1 (13)	$\min r$ (14)	$\nexists r$	$\exists r$
7	9	35		35
8	7	14		17
9	6	11	11	12
10	6	10		10
11	5	8	8	9

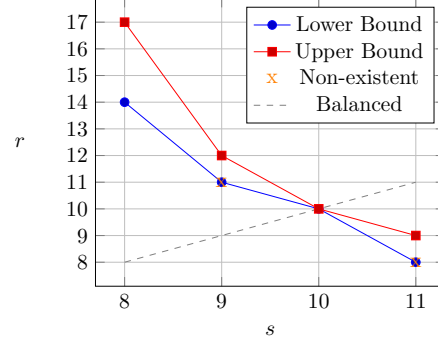


Fig. 4 Upper and lower bounds for $nc(s, 3, 2)$.

For completeness, the following proposition shows that the $nc(9, 3, 2, 12)$ covering also mentioned in [16] is optimal as well, since no $nc(9, 3, 2, 11)$ exists.

Theorem 13. *There is no $nc(9, 3, 2, 11)$ covering.*

Proof Suppose that there exists a $nc(9, 3, 2, 11)$ covering $\mathcal{C}$. According to Theorem 5, every element of U_9 is contained in at least $\sigma_1^{9,3,2} = 6$ sets and every 2-set of U_9 is contained in at least $\sigma_2^{9,3,2} = 3$ sets of $\mathcal{C}$ (see Table 2 as well). Therefore, $\mathcal{C}$ must contain at least $\sigma_1^{9,3,2} |U_9| = 6 \cdot 9 = 54$ elements of U_9 and $\sigma_2^{9,3,2} C(9, 2) = 3 \cdot 36 = 108$ 2-sets (since each of the 9 elements of U_9 must be contained in at least $\sigma_1^{9,3,2}$ sets, and each of the $C(9, 2)$ 2-sets of U_9 must be contained in $\sigma_2^{9,3,2}$ sets).

Per Theorem 2, the maximum size of a set in $\mathcal{C}$ is $m^{9,3,2} = 5$. In case some of the 11 sets of $\mathcal{C}$ were not of maximal size, the covering would contain at most $C(4, 2) + 10 \cdot C(5, 2) = 106$ 2-sets, since a set of size four contains $C(4, 2)$ 2-sets and the remaining 10 sets of size five contain at most $C(5, 2)$ 2-sets each. These 106 2-sets would not be enough to contain the necessary 108 2-sets. Thus $\mathcal{C}$ would have all its sets of maximal size and would contain exactly $m^{9,3,2} |\mathcal{C}| = 5 \cdot 11 = 55$ elements and $C(m^{9,3,2}, 2) |\mathcal{C}| = C(5, 2) \cdot 11 = 110$ 2-sets of U_9 . As a consequence, the two following statements hold.

1. Out of the 55 elements of the covering, 54 correspond to the minimum repetitions of each of the elements of U_9 . Therefore, one and only one of the elements of U_9 is contained into seven ($\sigma_1^{9,3,2} + 1$) sets of $\mathcal{C}$. All other elements are contained in exactly six ($\sigma_1^{9,3,2}$) elements of $\mathcal{C}$.
2. Out of the 110 2-sets of the covering, 108 correspond to the minimum three ($\sigma_2^{9,3,2}$) repetitions of each of the 2-sets of U_9 . Therefore, at least one of the 2-sets of U_9 is included in more than three sets of $\mathcal{C}$.

Finally, consider a 2-set $\{x, y\} \subset U_9$ such that (i) $\{x, y\}$ is included in more than three sets of $\mathcal{C}$, and (ii) x is included in exactly six sets of $\mathcal{C}$. Such 2-set exists, since there are 2-sets included in more than three sets (per point 2 above), but only one of the elements of U_9 is in more than six sets of $\mathcal{C}$ (per point 1 above). As such, x and y cannot both be included in more than six sets. Without loss of generality, we assume that x is included in six sets. Then, the six sets of $\mathcal{C}$ containing x must contain all 2-sets in the covering containing x . The amount of these 2-sets to be contained is at least $4 \cdot 1 + 3 \cdot 7 = 25$, since $\{x, y\}$ is contained

at least four times, and all other seven 2-sets of the form $\{x, z\}$ with $z \in u_9 \setminus \{x, y\}$ are contained at least three times. However, a set of size five containing x (say $\{x, z_1, z_2, z_3, z_4\}$) contains four of these 2-sets (namely $\{x, z_1\}$, $\{x, z_2\}$, $\{x, z_3\}$ and $\{x, z_4\}$). Therefore, the six sets containing x contain $6 \cdot 4 = 24$ such 2-sets, which are not enough to cover the 25 2-sets previously mentioned, reaching a contradiction. $\square$

The same line of reasoning used in the previous proof can be applied to show that a $nc(11, 3, 2, 8)$ covering does not exist (the proof is provided in [Appendix B](#)). These negative results are summarized in the third column of [Table 2](#) and shown with orange crosses in [Figure 4](#).

4.2 Class $(s, 3, 3)$ coverings

For this case, the application of [Theorem 5](#) to a $nc(s, 3, 3)$ covering whose largest set is of size m yields

$$\begin{aligned}\sigma_3^{s,3,3} &= 1 \\ \sigma_2^{s,3,3} &= 4 \\ \sigma_1^{s,3,3} &= \left\lceil \frac{4(s-1)}{m-1} \right\rceil.\end{aligned}\tag{15}$$

Based on these values, from [Theorem 7](#) we get the lower bound on the cardinality of the covering from the size of its largest set:

$$|\mathcal{C}| \geq \max \left\{ \left\lceil \frac{4(s-1)}{m-1} \right\rceil \frac{s}{m}, \frac{4C(s,2)}{C(m,2)}, \frac{C(s,3)}{C(m,3)} \right\}.\tag{16}$$

Moreover, [Theorem 10](#) is applicable to this case ($d = 3$), where the $d - 2$ sets become just one set. Taking such a set to be of the largest size m leads to

$$|\mathcal{C}| \geq \max \left\{ \left\lceil \frac{4(s-1)}{m-1} \right\rceil \frac{s-m}{s-m-4}, \frac{4C(s-m,2)}{C(s-m-4,2)}, \frac{C(s-m,3)}{C(s-m-4,3)} \right\} + 1.\tag{17}$$

For any number of elements s and a maximum set size m , the combination of equations (16) and (17) leads to the cardinality of the covering being at least

$$\begin{aligned}L_{s,m} = \max \left\{ \left\lceil \frac{4(s-1)}{m-1} \right\rceil \frac{s}{m}, \frac{4C(s,2)}{C(m,2)}, \frac{C(s,3)}{C(m,3)}, \right. \\ \left. \left\lceil \frac{4(s-1)}{m-1} \right\rceil \frac{s-m}{s-m-4} + 1, \frac{4C(s-m,2)}{C(s-m-4,2)} + 1, \frac{C(s-m,3)}{C(s-m-4,3)} + 1 \right\}.\end{aligned}$$

In this case, the maximum set size according to [Theorem 2](#) is $s - 7$. As such, the possible sizes of the largest set range from 3 to $s - 7$. The minimum among these bounds for all possible maximum sizes m is therefore a lower bound on the cardinality of a $nc(s, 3, 3)$ covering. That is,

$$|\mathcal{C}| \geq \min_{3 \leq m \leq s-7} L_{s,m}$$

The middle three columns of Table 3 show the values obtained for the number of repetitions of each individual element, the value of m for which the minimum in the previous expression is attained, and the corresponding lower bounds on the number of sets of the covering. Also, the blue line in Figure 5 shows these lower bounds.

In addition, the last column of Table 3 shows the sizes of the minimal coverings found (by means of the computational search) for the different values of s . To obtain them, the described algorithm was applied, although this time the sought coverings were of the maximal set sizes indicated in the table. One additional criterion for the selection of the set was added in this case: among the candidate sets that cover the maximum (or close to it in case this condition is slightly relaxed) number of 3-sets, the selection is made over those of minimal size. The intention is that non-completeness with other sets of the covering be as easy as possible.

It is concluded that the optimal balanced covering for degree $t = 3$ and order $d = 3$ would have s between 18 and 20. The computational search could only find a $nc(20, 3, 3, 20)$ covering, which is shown in Table A2. Although these results restrict the search to a somewhat narrow range of possibilities, whether there are $nc(18, 3, 3, 18)$ or $nc(19, 3, 3, 19)$ coverings is still an open question.

Table 3 Results of the search of the $(s, 3, 3)$ class of coverings.

s	σ_1 (15)	m	$\min r$ (16)	$\exists r$
10	18	3	120	120
11	14	4	42	85
12	11	5	36	69
13	12	5	32	56
14	11	6	26	43
15	10	7	22	36
16	9	8	20	30
17	8	9	20	26
18	9	9	18	22
19	8	10	16	20
20	8	11	16	19
21	8	11	16	19

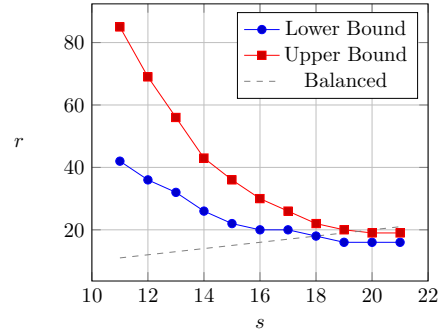


Fig. 5 Upper and lower bounds for $nc(s, 3, 3)$.

Moreover, we can obtain the dual for the $nc(19, 3, 3, 20)$ covering found. Table 4 includes the $nc(19, 3, 3, 20)$ covering whose sets correspond to the rows and the set elements with the check marks in each row. Then, the dual $nc(20, 3, 3, 19)$ covering is obtained by considering that its sets correspond to the columns and the set elements with the squares in each column. These coverings are complementary and thus can be used to create hardware implementations with a datapath of two widths. Ideally, an implementer would create 19 sub-functions for the non-linear layer and 20 for the linear layer. Note that sub-functions of the masked gadget tend to be more expensive than an extra share in the linear path.

Table 4 A $nc(19, 3, 3, 20)$ covering when taking check-marks by rows and a $nc(20, 3, 3, 19)$ covering when taking squares by columns.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	□	□	□	□	□	□	□	□	□
2	✓	□	✓	✓	□	✓	□	□	✓	□	✓	✓	□	□	✓	□	✓	✓	□
3	□	□	✓	✓	✓	✓	□	□	□	✓	□	□	✓	✓	✓	✓	✓	□	□
4	□	□	✓	✓	✓	□	✓	✓	□	□	✓	✓	□	✓	□	✓	□	✓	□
5	□	✓	✓	□	✓	□	□	□	✓	✓	✓	□	□	✓	□	□	□	✓	✓
6	✓	✓	✓	□	□	□	□	✓	□	✓	✓	✓	✓	✓	□	□	✓	□	□
7	□	✓	✓	□	□	✓	✓	□	✓	□	□	✓	□	✓	□	✓	✓	□	✓
8	✓	□	✓	□	✓	□	□	✓	✓	□	□	□	✓	□	□	✓	✓	✓	✓
9	□	□	✓	□	□	✓	✓	□	✓	□	✓	□	✓	□	✓	□	□	✓	✓
10	✓	✓	□	✓	□	□	✓	□	□	□	✓	□	✓	□	✓	✓	□	□	✓
11	□	✓	□	□	✓	✓	□	✓	✓	□	✓	✓	✓	□	✓	✓	□	□	□
12	□	□	□	✓	✓	□	✓	□	✓	✓	✓	✓	✓	□	□	□	✓	□	✓
13	✓	□	□	□	□	✓	✓	□	✓	✓	□	□	✓	✓	□	✓	□	✓	□
14	✓	□	□	□	✓	✓	✓	□	□	□	✓	□	□	✓	□	✓	□	□	✓
15	□	✓	□	✓	□	✓	□	✓	□	✓	✓	□	□	□	□	✓	✓	✓	✓
16	✓	✓	□	✓	✓	✓	□	□	□	□	□	✓	✓	✓	□	□	□	✓	✓
17	✓	□	□	✓	□	□	□	✓	✓	✓	□	✓	□	✓	✓	✓	□	□	✓
18	✓	✓	□	□	✓	□	✓	□	□	✓	□	✓	□	□	✓	✓	✓	✓	□
19	□	□	□	✓	□	□	✓	✓	✓	□	□	✓	✓	✓	✓	□	✓	✓	□
20	□	□	✓	✓	□	□	□	□	✓	✓	✓	□	□	□	□	□	□	□	✓

4.3 Class $(s, 4, 2)$ coverings

The application of [Theorem 5](#) to this case yields the following minimum number of repetitions of τ -sets in the covering for $1 \leq \tau \leq 4$.

$$\begin{aligned}
\sigma_4^{s,4,2} &= 1 \\
\sigma_3^{s,4,2} &= 3 \\
\sigma_2^{s,4,2} &= \left\lceil \frac{3(s-2)}{m-2} \right\rceil \\
\sigma_1^{s,4,2} &= \left\lceil \frac{\sigma_2^{s,4,2}(s-1)}{m-1} \right\rceil = \left\lceil \left\lceil \frac{3(s-2)}{m-2} \right\rceil \frac{s-1}{m-1} \right\rceil
\end{aligned} \tag{18}$$

From [Theorem 7](#) we get the following lower bound for the number of sets in the covering depending on the size of the largest set.

$$|\mathcal{C}| \geq \max \left\{ \left\lceil \left\lceil \frac{3(s-2)}{m-2} \right\rceil \frac{s-1}{m-1} \right\rceil \frac{s}{m}, \left\lceil \frac{3(s-2)}{m-2} \right\rceil \frac{C(s,2)}{C(m,2)}, \frac{3C(s,3)}{C(m,3)}, \frac{C(s,4)}{C(m,4)} \right\} \tag{19}$$

According to [Remark 1](#), the bound of [Theorem 10](#) does not apply in this case ($d = 2$).

The second and third columns of [Table 5](#) show the minimum number of repetitions of 2-sets and individual elements of U_s obtained. The fourth column shows the lower bounds obtained according to [Theorem 7](#) and [Equation \(19\)](#). These lower bounds are

obtained where m is of maximum size $m = s - t(d - 1) - 1$, that is, $m = s - 5$ in this case. The blue line in Figure 6 displays these bounds. In addition, the last column of Table 5 shows the sizes of the minimal coverings found (by means of the computational search) for the different values of s .

Table 5 Results of the search of the $(s, 4, 2)$ class of coverings.

s	σ_2	σ_1 (18)	$\min r$ (19)	$\exists r$
9	11	30	68	126
10	8	18	36	56
11	7	14	26	37
12	6	11	19	28
13	6	11	18	22
14	6	10	16	18
15	5	8	12	16
16	5	8	12	15

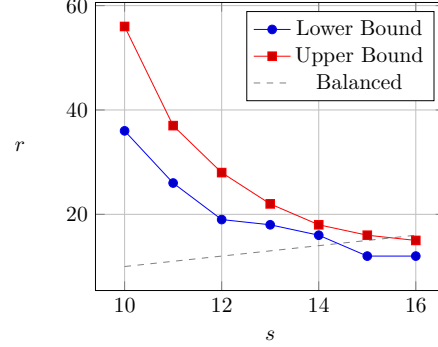


Fig. 6 Upper and lower bounds for $nc(s, 4, 2)$.

It is concluded that the optimal balanced covering for degree $t = 4$ and order $d = 2$ would have s equal to 15 or 16. The computational search could only find a $nc(16, 4, 2, 16)$ covering, which is shown in Table A3. Whether there is a $nc(15, 4, 2, 15)$ coverings remains an open question.

5 Open problems

Several directions might be explored in future works. For instance, the conditions for sufficiency of sizes and number of sets of Proposition 6 and Proposition 9 may be combined and explored from the perspective of integer programming, since a minimal covering represents an optimal solution to the minimization problem where the number of sets is the objective and the results in these propositions are linear restrictions. Also, the representation of an NCSC as an ensemble of sub-coverings sharing sets may be explored to obtain existence conditions and, eventually, constructive methods.

In terms of the computational search, it can be expanded to higher orders of security. A challenge faced by the proposed algorithm is its extensive memory use. The proposed algorithm allocates all candidate sets, what may become unfeasible for larger values of s , t and d .

Also, ad hoc proofs of non-existence may be sought for the yet undetermined cases described in Section 4. Namely, whether $nc(18, 3, 3, 18)$, $nc(19, 3, 3, 19)$ and $nc(15, 4, 2, 15)$ coverings exist. Moreover, it is also an open question whether a $nc(8, 3, 2)$ covering with less than 17 sets exists.

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References

- [1] Mangard, S., Oswald, E., Popp, T.: Power Analysis Attacks: Revealing the Secrets of Smart Cards. Springer, Boston MA, USA (2007)
- [2] Kocher, P., Jaffe, J., Jun, B.: Differential power analysis. In: Proceedings of the 19th Annual International Cryptology Conference (CRYPTO), pp. 388–397. Springer, Berlin, Heidelberg (1999)
- [3] Chari, S., Jutla, C.S., Rao, J.R., Rohatgi, P.: Towards sound approaches to counteract power-analysis attacks. In: Proceedings of the 19th Annual International Cryptology Conference (CRYPTO), pp. 398–412. Springer, Berlin, Heidelberg (1999)
- [4] Goubin, L., Patarin, J.: DES and Differential Power Analysis The “Duplication” Method. In: Proceedings of the 1999 Workshop on Cryptographic Hardware and Embedded Systems (CHES), pp. 158–172. Springer, Berlin, Heidelberg (1999)
- [5] Messerges, T.S.: Using Second-Order Power Analysis to Attack DPA Resistant Software. In: Proceedings of the 2000 Workshop on Cryptographic Hardware and Embedded Systems (CHES), pp. 238–251. Springer, Berlin, Heidelberg (2000)
- [6] Mangard, S., Popp, T., Gammel, B.M.: Side-channel leakage of masked CMOS gates. In: The Cryptographers’ Track at the RSA Conference (CT-RSA), pp. 351–365. Springer, Berlin, Heidelberg (2005)
- [7] Mangard, S., Pramstaller, N., Oswald, E.: Successfully attacking masked AES hardware implementations. In: Proceedings of the 2005 Workshop on Cryptographic Hardware and Embedded Systems (CHES), pp. 157–171. Springer, Berlin, Heidelberg (2005)
- [8] Nikova, S., Rechberger, C., Rijmen, V.: Threshold implementations against side-channel attacks and glitches. In: Proceedings of the 8th International Conference on Information and Communications Security (ICICS), pp. 529–545. Springer, Berlin, Heidelberg (2006)
- [9] Bilgin, B., Gierlichs, B., Nikova, S., Nikov, V., Rijmen, V.: Higher-order threshold implementations. In: Proceedings of the 20th International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), pp. 326–343. Springer, Berlin, Heidelberg (2014)
- [10] Reparaz, O.: A note on the security of Higher-Order Threshold Implementations. Cryptology ePrint Archive, Paper 2015/001 (2015)

- [11] Reparaz, O., Bilgin, B., Nikova, S., Gierlichs, B., Verbauwhede, I.: Consolidating Masking Schemes. In: Proceedings of the 2015 Annual International Cryptology Conference (CRYPTO), pp. 764–783. Springer, Berlin, Heidelberg (2015)
- [12] Moos, T., Moradi, A., Schneider, T., Standaert, F.-X.: Glitch-Resistant Masking Revisited: or Why Proofs in the Robust Probing Model are Needed. IACR Transactions on Cryptographic Hardware and Embedded Systems **2019**(2), 256–292 (2019)
- [13] Petrides, G.: On non-completeness in threshold implementations. In: Proceedings of ACM Workshop on Theory of Implementation Security Workshop (TIS), pp. 24–28 (2019)
- [14] Wegener, F., Moradi, A.: Yet Another Size Record for AES: A First-Order SCA Secure AES S-Box Based on $GF(2^8)$ Multiplication. In: Smart Card Research and Advanced Applications (CARDIS), pp. 111–124. Springer, Cham (2019)
- [15] Arribas, V., Bilgin, B., Petrides, G., Nikova, S., Rijmen, V.: Rhythmic KECCAK: SCA security and low latency in HW. IACR Trans. Cryptogr. Hardw. Embed. Syst., 269–290 (2018)
- [16] Grosso, V., Lara-Nino, C.A.: Leveraging the resources of modern FPGAs in the design of high-performance masked architectures. In: Proceedings of the 18th Reunión Española Sobre Criptología Y Seguridad de la Información (RECSI), pp. 1–6. Universidad de León, León, Spain (2024)
- [17] Grosso, V., Lara-Nino, C.A.: Overshooting the Threshold: $(td+n)$ -Masking. Cryptology ePrint Archive, Paper 2025/1936 (2025). <https://eprint.iacr.org/2025/1936>
- [18] Piccione, E., Andreoli, S., Budaghyan, L., Carlet, C., Dhooghe, S., Nikova, S., Petrides, G., Rijmen, V.: An optimal universal construction for the Threshold Implementation of bijective S-Boxes. IEEE Transactions on Information Theory **69**(10), 6700–6710 (2023)
- [19] Baksi, A., Guilley, S., Shrivastwa, R.-R., Takarabt, S.: From Substitution Box to Threshold. In: Chattopadhyay, A., Bhasin, S., Picek, S., Rebeiro, C. (eds.) Progress in Cryptology – INDOCRYPT 2023, pp. 48–67. Springer, Cham (2024)
- [20] Carlet, C.: Boolean Functions for Cryptography and Coding Theory. Cambridge University Press, Cambridge, UK (2021)
- [21] Cramer, R., Damgård, I.B., Nielsen, J.B.: Secure Multiparty Computation and Secret Sharing. Cambridge University Press, Cambridge (2015)
- [22] Fidalgo Rodríguez, O.: On the search of non-complete set coverings for the design of protected substitution layers. Master’s thesis, Universitat Oberta de Catalunya

Appendix A Minimal balanced coverings obtained

The three following tables include the balanced coverings of the smallest cardinality that were found using the search algorithm described in [Section 4](#).

	1	2	3	4	5	6	7	8	9	10
1	✓	✓	✓	✓	✓	✓	□	□	□	□
2	✓	□	□	✓	□	✓	□	✓	✓	✓
3	✓	✓	□	□	✓	□	□	✓	✓	✓
4	✓	□	✓	□	✓	✓	✓	□	□	✓
5	□	✓	□	✓	✓	□	✓	□	✓	✓
6	✓	□	✓	✓	□	□	✓	✓	✓	□
7	□	✓	✓	□	□	✓	✓	□	✓	✓
8	□	✓	✓	□	✓	✓	□	✓	✓	□
9	□	□	✓	✓	✓	□	✓	✓	□	✓
10	✓	✓	□	✓	□	✓	✓	✓	□	□

Table A1 A $nc(10, 3, 2, 10)$ covering.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	□	□	□	□	□	□	□	□	□
2	□	✓	✓	✓	✓	□	□	✓	□	✓	□	□	✓	□	✓	✓	□	✓	✓	□
3	✓	✓	□	✓	□	□	□	✓	□	✓	✓	✓	□	✓	□	□	□	✓	✓	✓
4	□	✓	□	✓	□	✓	✓	✓	□	✓	□	□	□	✓	✓	✓	✓	□	□	✓
5	□	□	□	✓	✓	✓	□	✓	✓	□	✓	✓	✓	✓	□	□	✓	□	□	□
6	✓	□	□	✓	□	✓	✓	□	□	✓	✓	✓	✓	□	✓	✓	□	✓	□	□
7	✓	□	✓	✓	□	✓	□	□	✓	✓	□	□	□	□	□	✓	✓	✓	✓	✓
8	□	✓	✓	✓	□	□	□	□	✓	✓	✓	✓	✓	□	✓	□	✓	□	□	✓
9	✓	□	✓	✓	✓	□	✓	□	□	✓	□	✓	□	✓	✓	□	✓	□	✓	□
10	□	□	□	✓	✓	□	✓	□	✓	✓	✓	□	✓	✓	□	✓	□	□	✓	✓
11	✓	□	✓	✓	□	✓	✓	✓	□	□	□	□	✓	✓	✓	□	□	□	□	✓
12	✓	✓	□	✓	✓	□	□	✓	□	✓	□	✓	□	✓	✓	✓	✓	✓	□	□
13	□	□	✓	□	✓	□	✓	✓	□	□	✓	✓	□	□	□	✓	✓	✓	□	✓
14	□	✓	□	✓	□	✓	✓	□	✓	□	□	✓	□	□	✓	✓	□	✓	✓	□
15	✓	□	□	□	✓	□	□	✓	□	✓	□	✓	✓	□	✓	□	✓	□	✓	□
16	✓	✓	✓	✓	✓	✓	□	□	□	□	□	✓	✓	✓	□	✓	□	□	□	✓
17	□	□	✓	□	✓	✓	□	□	□	□	✓	□	□	✓	✓	□	□	□	✓	✓
18	□	✓	□	✓	□	□	✓	□	✓	□	□	✓	✓	□	□	□	✓	□	✓	□
19	✓	□	□	□	□	□	□	✓	✓	□	□	✓	□	□	✓	✓	□	□	✓	□
20	□	□	□	□	□	□	✓	✓	□	□	✓	□	✓	✓	□	✓	✓	□	✓	□

Table A2 A $nc(20, 3, 3, 20)$ covering.

Appendix B There is no $nc(11, 3, 2, 8)$ covering

Theorem 14. *There is no $nc(11, 3, 2, 8)$ covering.*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	□	□	□	□	□
2	✓	□	✓	□	✓	□	✓	✓	✓	□	✓	✓	✓	✓	□	✓
3	□	✓	□	✓	□	□	✓	✓	✓	✓	✓	✓	□	✓	✓	✓
4	✓	□	□	✓	✓	✓	✓	✓	□	✓	□	□	✓	✓	✓	✓
5	✓	✓	✓	□	□	✓	✓	□	□	✓	✓	✓	✓	✓	✓	□
6	□	✓	✓	✓	✓	✓	✓	□	✓	□	□	✓	✓	□	✓	✓
7	✓	✓	✓	✓	□	✓	□	✓	✓	□	□	✓	✓	✓	✓	□
8	✓	□	✓	✓	✓	□	□	□	✓	✓	✓	✓	✓	□	✓	✓
9	□	✓	✓	□	✓	✓	□	✓	□	✓	✓	✓	□	✓	✓	✓
10	✓	✓	□	✓	✓	✓	□	□	✓	✓	✓	□	✓	✓	□	✓
11	□	□	□	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	□	✓	□
12	✓	✓	✓	✓	✓	□	✓	✓	□	✓	□	✓	✓	□	□	✓
13	✓	□	✓	✓	✓	✓	✓	✓	✓	✓	□	✓	□	□	□	✓
14	✓	✓	✓	□	✓	✓	✓	✓	✓	□	✓	□	□	□	✓	✓
15	□	✓	✓	✓	✓	□	✓	✓	□	□	✓	✓	✓	✓	□	□
16	□	□	□	□	✓	□	□	□	✓	□	✓	✓	✓	✓	✓	□

Table A3 A $nc(16, 4, 2, 16)$ covering.

Proof The proof of this theorem is analogous to the proof of [Theorem 13](#). Suppose that there exists a $nc(11, 3, 2, 8)$ covering $\mathcal{C}$. According to [Theorem 5](#), every element of U_{11} is contained in at least $\sigma_1^{11,3,2} = 5$ sets and every 2-set of U_{11} is contained in at least $\sigma_2^{11,3,2} = 3$ sets of $\mathcal{C}$ (see [Table 2](#) as well). Therefore, $\mathcal{C}$ must contain at least $\sigma_1^{11,3,2} |U_{11}| = 5 \cdot 11 = 55$ elements of U_{11} and $\sigma_2^{11,3,2} C(11, 2) = 3 \cdot 55 = 165$ 2-sets (since each of the 11 elements of U_{11} must be contained in at least $\sigma_1^{11,3,2}$ sets, and each of the $C(11, 2)$ 2-sets of U_{11} must be contained in $\sigma_2^{11,3,2}$ sets).

Per [Theorem 2](#), the maximum size of a set in $\mathcal{C}$ is $m^{11,3,2} = 7$. In case some of the 8 sets of $\mathcal{C}$ were not of maximal size, the covering would contain at most $C(6, 2) + 7 \cdot C(7, 2) = 15 + 7 \cdot 21 = 162$ 2-sets, since a set of size six contains $C(6, 2)$ 2-sets and the remaining 7 sets of size seven contain at most $C(7, 2)$ 2-sets each. These 162 2-sets would not be enough to contain the necessary 165 2-sets.

Thus, $\mathcal{C}$ would have all its sets of maximal size and would contain exactly $m^{11,3,2} |\mathcal{C}| = 7 \cdot 8 = 56$ elements and $C(m^{11,3,2}, 2) |\mathcal{C}| = C(7, 2) \cdot 8 = 21 \cdot 8 = 168$ 2-sets of U_{11} . As a consequence, the two following statements hold.

1. Out of the 56 elements of the covering, 55 correspond to the minimum repetitions of each of the elements of U_{11} . Therefore, one and only one of the elements of U_{11} is contained into six $(\sigma_1^{11,3,2} + 1)$ sets of the covering. All other elements are contained in exactly five $(\sigma_1^{11,3,2})$ elements of the covering.
2. Out of the 168 2-sets of the covering, 165 correspond to the minimum three $(\sigma_2^{11,3,2})$ repetitions of each of the 2-sets of U_{11} . Therefore, at least one of the 2-sets of U_{11} is included in more than three sets of $\mathcal{C}$.

Finally, consider a 2-set $\{x, y\} \subset U_{11}$ such that (i) $\{x, y\}$ is included in more than three sets of $\mathcal{C}$, and (ii) x is included in exactly five sets of $\mathcal{C}$. Such 2-set exists, since there are 2-sets included in more than three sets (per point 2 above), but only one of the elements of U_{11} is included in more than five sets of $\mathcal{C}$ (per point 1 above). As such, x and y cannot both be included in more than five sets. With no loss of generality, we assume x is included in five sets. Then, the five sets of the covering containing x must contain all 2-sets in the

covering containing x . The amount of these 2-sets to be contained is at least $4 \cdot 1 + 3 \cdot 9 = 31$, since $\{x, y\}$ is contained at least four times, and all other nine 2-sets of the form $\{x, z\}$ with $z \in u_{11} \setminus \{x, y\}$ are contained at least three times. However, a set of size seven containing x (say $\{x, z_1, z_2, \dots, z_6\}$) contains six of these 2-sets (namely $\{x, z_1\}, \{x, z_2\} \dots \{x, z_6\}$). Therefore, the five sets containing x contain $5 \cdot 6 = 30$ such 2-sets, which are not enough to cover the 31 2-sets previously mentioned, reaching a contradiction. $\square$